MILITARY SPECIFICATION

COUPLERS, PASSIVE, FIBER OPTIC, GENERAL SPECIFICATION FOR (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and characteristics of passive fiber optic couplers intended to optically interconnect three or more optical fibers. Directional and bidirectional couplers are covered by this document. Couplers are compatible with and suitable for interconnection to cable fibers as specified in DOD-C-85045. They are also compatible with splices as specified in DOD-S-24623, and connectors as specified in MIL-C-28876.

1.2 Classification. Coupler type refers to the method of interfacing the coupler to external fiber optic cables for system applications. Couplers shall be of the following types.

Type I: Couplers with fiber optic cable connectors as an integral part of the coupler assembly housing.

Type II: Couplers with integral fiber optic cable "pig-tails" as part of the coupler assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-766	-	Steel Plate, Sheet, and Strip-Corrosion Resisting.

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MIL-S-901	-	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-S-5002	-	Surfaces Treatments and Inorganic Coatings for Metal Surfaces of Weapons Systems.
MIL-I-17214	-	Indicator, Permeability, Low-mu (Go-no-go).
MIL-L-17331	-	Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-L-23699	-	Lubricating Oil, Aircraft Turbine Engines, Synthetic Base.
DOD-C-24621/1	-	Coupler, Fiber Optic Cable, Cable Splitter, Passive Glass-Connectorized Output; Type I 820 to 910 Nanometers Wavelength Range, 2 X 2 Symmetrical Transmission Star, 50/125 Micrometer Fiber, Graded Index Multi-Mode Bidirectional.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Sea Systems Command, ATTN: SEA 55Z3, Department of the Navy, Washington, DC 20362, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DOD-C-24621/2	-	Coupler, Fiber Optic Cable, Cable Splitter, Passive Glass-Pigtailed Output; Type II 820 to 910 Nanometers Wavelength Range, 4 X 4 Symmetrical Transmission Star, 50/125 Micrometer Fiber, Graded Index Multi-Mode Bidirectional.
DOD-S-24622	-	Sources, LED, Fiber Optic, General Specification For (METRIC).
MIL-C-28876	-	Connectors, Fiber Optic, Environmental Resisting (For Navy Shipboard Application) General Specification for.
MIL-C-55330	-	Connectors, Electrical and Fiber Optic, Packaging of.
DOD-C-85045	-	Cables, Fiber Optics, General Specification for (METRIC).
DOD-C-85045/1	-	Cable, Fiber Optics, type I, Class 2, Composition A, Attenuation Range A, Single Fiber For.

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STANDARDS

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.
MIL-STD-1344	-	Test Methods for Electrical Connectors.
DOD-STD-1678	-	Fiber Optics Test Methods and Instrumentation.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual coupler requirements shall be as specified herein, and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Fiber optic coupler parts furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.3 Material. The coupler, coupler connector caps, covers, storage receptacles or other protective accessory hardware shall be constructed of material as specified herein or in the specification sheet. The material used in the coupler shall be as follows:

a. Resistant to corrosion as specified in MIL-STD-454, requirement 15.

b. Inert to fungus (nonnutritive) as specified in MIL-STD-454, requirement 4.

c. Nontoxic, nonflammable, and nonexplosive as specified in MIL-STD-454, when exposed to heat, flame, or chemicals.

d. Nonmagnetic under normal circumstances (with maximum microns (μ) less than or equal to 2.0 as determined by measurement with equipment conforming to MIL-I-17214).

e. Devoid of elemental mercury and asbestos.

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f. Manufactured to good workmanship quality as specified in MIL-STD-454, requirement 9.

3.3.1 Steel parts. All exposed corrosion-resistant steel parts (as specified in QQ-S-766) of the coupler assembly shall have a passivated finish which permits the attainment of a surface finish condition in accordance with high workmanship standards and shall be compatible with external coatings or platings of the type and color as specified in MIL-S-5002.

3.3.2 Dissimilar metal junctions. Where dissimilar metals are used in intimate contact with one another, suitable protection against electrolytic corrosion shall be provided as specified in MIL-STD-889.

3.3.3 Solvents, adhesives, and cleaning agents. If any chemicals or cements are used in bonding of internal coupler parts or the fiber/contact securing-polishing processes, incompatibility shall not exist between the materials employed such that degradation of these materials shall result from in-service use. Any epoxy and adhesives used in coupler construction shall conform to requirement 23 of MIL-STD-454. Additionally, the coupler manufacturer shall supply a list of recommended solvents, adhesives, and cleaning agents.

3.3.4 Fiber material. The optical material of the couplers shall be compatible with the cable as specified in DOD-C-85045/1. The types of fiber are as follows:

a. Step index: Glass cladding, glass core.

b. Step index: Plastic cladding, glass core.

c. Step index: Plastic cladding, plastic core.

d. Graded index: Glass.

3.4 Design and construction. The coupler shall be mechanically and optically compatible with those sizes and types of fiber optic cables as specified in DOD-C-85045. Where cables or connectors are employed as part of the coupler assembly on the outside of the coupler enclosure, the cables or connectors are expected to meet the applicable fiber optic cable or connector requirements as specified in DOD-C-85045 or MIL-C-28876, as applicable.

3.4.1 Component design and construction. Couplers and associated accessories constructed to this specification shall withstand normal handling incident to installation and in-service maintenance.

3.4.1.1 Primary design considerations. Primary design shall be as follows:

a. Reliable for operation under conditions of temperature, salt spray, moisture, and vibration as described herein and in accordance with requirement 35 of MIL-STD-454.

b. Lightweight and compact.

c. Simple design, minimum number of maintenance parts or moveable parts, and maximum ease of repair.

d. Maximum compatibility shall exist between couplers, connector receptacles, and interconnecting cables without modification thereto or use of special adapters.

e. Maximum interchangeability of parts; interchangeability shall conform to requirement 7 of MIL-STD-454.

f. Maximum ease of installation.

g. Satisfactory identification for ease of requisitioning.

h. Maximum protection for operating and maintenance personnel.

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i. Minimum operating and maintenance cost.

j. Methods for shipboard mounting in permanent position.

3.4.2 Seals. Grommets, O-rings, interfacial seals, boots, gaskets or other sealing devices, as needed by the particular coupler design, shall accomplish their intended purpose and meet all test requirements as specified herein.

3.4.3 Optical junction sealing. Optical junctions shall be sealed against moisture and contamination as specified herein.

3.4.4 Protective covers. Coupler-connector receptacles or "pig-tail" cable connectors shall be provided with a protective cover to prevent exposure and damage from ordinary handling, moisture, and dust when the mating connector is not attached. Protective covers shall be molded or machined from plastic or nonferrous material suitable for fitting the appropriate coupler part. Each plug or receptacle connector shall be fitted with a slide-on or screw-on protective cap on the front end of each assembly, as applicable, prior to unit packaging to protect contacts and seal surfaces from entry of foreign matter, and to protect the surfaces from mechanical damage.

3.5 Personnel safety requirements. Couplers shall not present a hazard or permit cause for injury to personnel during normal assembly (if performed by the user), installation, or usage. Type II couplers (requiring splices for installation) shall be supplied with instruction sheets which contain information describing appropriate safety features and precautions, and safe methods of making the interconnection splices. Material and equipment used shall have no adverse effect on the health of personnel when used properly for the purpose intended. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting activity agency. The manufacturer's instruction sheets shall emphasize precautions similar to those stated below. Sample statement addresses special areas: "Care will be taken when connecting and handling the sharp and very fine diameter optical fibers to prevent skin puncture or contact of fiber with the eye area. Also, direct viewing of the optical terminal face of a terminated cable or coupler output port while it is propagating optical energy is not recommended unless prior assurance has been obtained as to the safe power output level of the terminal or port."

3.6 Optical performance.

3.6.1 Insertion loss. For each coupler specimen tested in accordance with 4.6.1, the measured insertion loss between the input port J to the output port I (including induced attenuation due to environmental stress) shall not exceed the limit as specified (see 3.1). This insertion loss also includes that attenuation due to the input and output connector receptacles in the case of type I couplers; or that due to pigtail extension leads in the case of type II couplers.

3.6.2 Signal discontinuity. When tested in accordance with 4.6.2, discontinuities shall not occur in the coupler's optical circuit signal. A discontinuity is considered a reduction in signal strength of 2 dB or more for a duration of 20 nanoseconds (ns) or more.

3.6.3 Analog modulation. For each coupler subjected to time-dependent environmental or mechanical testing (see 4.6.3), the resultant modulation observed in an optical circuit signal, bandpass limited to between 4 Hz and 40 kHz, shall not exceed 5 percent of the steady-state signal level.

3.6.4 Pulse distortion. When tested in accordance with 4.6.4, the rise and fall times of the transmitted optical pulses shall be at least as long as, but not more than 1 ns longer than the rise and fall time of the pulses incident on the optical circuit of the coupler. Additionally, the duration of the transmitted optical pulses shall be within 2 ns of the duration of the incident optical pulses comprising the reference signal.

3.6.5 Ambient optical pickup. When tested in accordance with 4.6.5, the optical power of the light from the nonenergized optical ports (after accounting for cable and optical junction losses between the coupler device and the detector) shall be less than -70 dBm.

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3.6.6 Transmission matrix. Each coupler shall be tested for quality of performance according to the test configuration described by 4.6.6. Each value of P_{IJ} must be determined, and found to agree with the value specified (see 3.1) (within the allotted tolerance). The quantities P_{IJ} of this matrix when combined and substituted in the proper mathematical expression also define all the port-to-port and total insertion loss values for a particular coupler.

3.6.7 Optical power - angular distribution. The output characteristics of the coupler shall match, within tolerance, the input characteristics of an injected fiber in terms of signal intensity as a function of angular direction. The comparison shall be made in accordance with the conditions and procedures of 4.6.7.

3.6.7.1 Input pattern measurement. The angular radiation pattern injected into the coupler input port shall be determined in accordance with 4.6.7.1. Quantitative measurements of this input pattern shall agree with government approved manufacturer's specification. Also, the quantitative measurements obtained (see 4.6.7) shall define the angular reference pattern for comparison tests as specified herein.

3.6.7.2 Coupler radiation pattern measurement. The angular radiation output pattern from the coupler shall be determined in accordance with 4.6.7.2. Quantitative measurements of this radiation pattern shall conform to the measurements obtained for angular profile as specified in 4.6.7.1. The normalized output power distribution (energy versus angle) must not deviate by more than 10 percent at any angle from that power measurement at the same angle in the angular distribution recorded for the corresponding input pattern (as measured without coupler inserted in test apparatus).

3.7 Environmental requirements.

3.7.1 Accelerated aging. When tested in accordance with 4.7.1, couplers shall not exhibit visual evidence of dimensional change, opening of seals, cracking or crazing of components or finishes, identification marking impairment, fusion or seizure of mating parts, leakage of waterproofing compounds or other effects detrimental to coupler operation. Evidence of adhesive degradation shall not be present. Index matching fluids, if used, shall not be replenished after temperature exposure. (No revitalization of coupler condition shall be allowed before exposure to remaining test sequences.) A posttest visual inspection shall reveal no jacket softening, surface damage, or identification marking impairment. Posttest coupler output cable outer diameter shall remain within ±10 percent of the pretest cable outer diameter. The insertion loss shall not exceed the allowable limits as specified (see 3.1) before and after this exposure. Also, the coupler must meet the requirements for ambient optical pickup (see 3.6.5) and pulse distortion (see 3.6.4).

3.7.2 Ozone exposure. When tested in accordance with 4.7.2, seals shall show no evidence of swelling which may degrade environmental isolation. The insertion loss shall not exceed the allowable limits as specified (see 3.1) before and after this exposure.

3.7.3 Fluid immersion. When tested in accordance with 4.7.3, a posttest visual inspection shall reveal no cracks, splits, gumminess or voids in the coupler assembly. The coupler cable outer diameter shall not experience a change greater than ±10 percent as referenced to the pretest coupler pigtail cable diameter. Visual inspection of the test coupler shall reveal no swelling or softening of material, no loss of sealing capability or identification marking and no discoloration or other effects detrimental to the intended use of these optical components. The insertion loss shall not exceed the allowable limits as specified (see 3.1) before and after this exposure.

3.7.4 Humidity. When tested in accordance with 4.7.4 for accelerated environment exposures of high humidity and cyclic temperatures, internal and external parts shall not swell or otherwise degrade such that coupler performance is impaired. A posttest visual inspection shall reveal no component softening, surface damage or identification marking impairment. Posttest coupler output cable outer diameter shall remain within ±10 percent of the pretest pigtail cable outer diameter. The insertion loss shall not exceed the allowable limits as specified (see 3.1) before and after this exposure.

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3.7.5 Water pressure. When tested in accordance with 4.7.5, visual inspection of the test coupler shall reveal no penetration of indicator dye into the sealed region of the couplers. The insertion loss shall not exceed the allowable limits as specified (see 3.1) before and after this exposure.

3.7.6 Identification marking durability. When tested in accordance with 4.7.6, the coupler identification marking shall remain legible and intact.

3.7.7 Impact drop (impact resistance). When tested in accordance with 4.7.7, couplers and associated connectors shall not be visibly damaged or otherwise rendered unfit for operational use. Also, the insertion loss shall not exceed the allowable limits as specified (see 3.1) before and after each event of this test.

3.7.8 Axial cable loading.

3.7.8.1 Axial cable loading/tension (type II couplers only). When tested in accordance with 4.7.8.1, the insertion loss shall not exceed the allowable limits as specified (see 3.1) at any time during and after this exposure. The posttest visual inspection shall reveal no cracking, splitting, regions of localized tearing, or deformation.

3.7.8.2 Axial cable loading/compression (type II couplers only). When tested in accordance with 4.7.8.2, axial movement of the pigtail cable relative to the enclosure shall be no more than half the cable diameter. A posttest inspection of the coupler shall reveal no physical damage to the seals. The insertion loss shall not exceed the allowable limits as specified (see 3.1) at any time during and after this exposure.

3.7.9 Enclosure bending moment. When tested in accordance with 4.7.9, couplers shall sustain no permanent deformation. The insertion loss shall not exceed the allowable limits as specified (see 3.1) at any time during and after this exposure.

3.7.10 Cable twist (type II couplers only). When tested in accordance with 4.7.10, a posttest visual inspection of the coupler and cable assembly shall reveal no loss of environmental sealing, splitting, or other impairment due to the torsional bending exposure. The insertion loss shall not exceed the allowable limits as specified (see 3.1) at any time during and after this exposure.

3.7.11 Cable seal flexing (type II couplers only). When tested in accordance with 4.7.11, coupler strain relief mechanisms shall prevent loss of environmental sealing or other damage which may impair the coupler operation. The insertion loss shall not exceed the allowable limits as specified (see 3.1) at any time during and after this exposure.

3.7.12 Mechanical shock. When tested in accordance with 4.7.12, coupler assemblies shall not be damaged and there shall be no loosening of parts. The insertion loss shall not exceed the allowable limits as specified (see 3.1) before and after this exposure.

3.7.13 Thermal shock. When tested in accordance with 4.7.13, a posttest visual inspection shall reveal no softening, surface damage, or identification marking impairment. For the case of type II couplers, posttest pigtail cable outer diameter shall remain within ±10 percent of the pretest cable diameter. The test couplers shall reveal no evidence of part dimensional change, leakage of water-proofing compounds, or other apparent loss of sealing capability, coupling-thread binding or other evidence of mating or unmating incapability, and no other damage detrimental to the operation of the coupler. The insertion loss shall not exceed the allowable limits as specified (see 3.1) before and after this exposure.

3.7.14 Operating temperature. When tested in accordance with 4.7.14, a posttest visual inspection of the coupler shall reveal no defects or damage. The coupler body and pigtail cable leads shall show no evidence of temperature susceptibility in regards to this test. The coupler shall survive and show no signs of mating or unmating incompatibility, leakage or waterproof sealing degradation, swelling, stretching or shrinking, or surface identification marking impairment. The insertion loss shall not exceed the allowable limits as specified (see 3.1) during this exposure.

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3.7.15 Vibration. When tested in accordance with 4.7.15, the coupler shall exhibit no loosening of parts, relative motion between parts, or other damage which can produce physical distortion or wear and may result in fatigue of the mechanical parts or failure of the coupler. The insertion loss shall not exceed the allowable limits as specified (see 3.1) before, during, and after this exposure. Also, the coupler must meet the requirements for discontinuity, analog modulation, and the pulse distortion (see 3.6.2, 3.6.3, and 3.6.4) during this exposure.

3.7.16 Salt spray (corrosion). When tested in accordance with 4.7.16, no visible evidence of corrosion shall be seen on the external or associated coupler parts. The insertion loss shall not exceed the allowable limits as specified (see 3.1) before and after this exposure.

3.7.17 Flammability. When couplers are tested in accordance with 4.7.17, burning after removal of the applied flame shall extinguish within the time specified by test condition B, method 1012 (see table I) of MIL-STD-1344. Any observed after-glow shall cease within the time specified by test condition B, method 1012 (see table II) of MIL-STD-1344. Also, there shall be no dripping which will cause ignition of flammable material placed below the flame's contact point with coupler. There shall be no violent burning or explosive type fire during and after flame application. In the case of type II couplers, burning and after-glow of pigtail-fiber cables tested under method 5010 of DOD-STD-1678 shall conform to the requirements of DOD-C-85045.

3.8 Dimensions. The dimensions and dimensional tolerances for these couplers and associated accessories shall conform to the limits as specified (see 3.1).

3.9 Mass. The mass of the couplers and associated accessories shall conform to the limits when specified (see 3.1).

3.10 Color. The external color of the couplers shall be as specified (see 6.2.1).

3.11 Marking. Unless otherwise specified (see 6.2.1), the finished coupler shall be identified by a marking applied to the outer surface of the coupler or visible through the outer surface. The characteristics and method of identification marking shall conform to the requirements of MIL-STD-1285. The identification marking characters shall be 2 millimeters in height unless otherwise specified (see 3.1). The identification marking shall be permanent and legible in accordance with the marking requirements of MIL-STD-1285. In particular, the identification marking shall also include the words "Fiber Optic Coupler". Identification marking shall be applied with the vertical axes of the printed characters perpendicular to the longest dimension of the coupler housing. All characters shall be expected to remain legible for the 20 year expectancy lifetime of the coupler. Except as otherwise specified, the coupler accessory parts shall also be identified by a legible and permanent marking applied in accordance with MIL-STD-1285.

3.12 Workmanship. Couplers and accessories shall be free of manufacturing flaws that would degrade performance, inhibit proper connection to interfacing elements, and otherwise yield an inferior product. All details of workmanship shall be in accordance with high grade fiber optic technology manufacturing practice. The following are minimal considerations and are not intended to exclude other pertinent workmanship requirements deemed necessary by the contractor.

a. Outer surface shall be free of rust, burnt areas, abrasions, holes, roughhead areas, bulges, thin spots, and indentations.

b. Inner parts shall be free of fractures, holes, discontinuities, bulges, and thin spots.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Test equipment and inspection facilities. The manufacturer shall insure that test and inspection facilities of sufficient accuracy, quality, and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of DOD-STD-1678.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory satisfactory to the Naval Sea Systems Command (NAVSEA) (see 6.3) on sample units produced with equipment and procedures normally used in production.

4.4.1 Sample. The number of sample units comprising a sample of couplers to be subjected to qualification inspection shall be six. The sample shall be taken at random from a production run and shall be produced with equipment and procedures normally used in production.

4.4.2 Inspection routine. Sample units shall be subjected to the qualification inspection specified in table I in the order shown.

TABLE I. Qualification inspection.

Inspection	Requirement paragraph	Test method paragraph
Group I		
Visual and mechanical inspection	3.3, 3.4, 3.5, 3.8, 3.9, 3.10, 3.11, 3.12	---
Insertion loss	3.6.1	4.6.1
Pulse distortion	3.6.4	4.6.4
Ambient optical pickup	3.6.5	4.6.5
Transmission matrix	3.6.6	4.6.6
Optical power-angular distribution	3.6.7	4.6.7
Input pattern measurement	3.6.7.1	4.6.7.1
Coupler radiation pattern measurement	3.6.7.2	4.6.7.2
Group II (2 samples)		
Accelerated aging	3.7.1	4.7.1
Ozone exposure	3.7.2	4.7.2
Fluid immersion	3.7.3	4.7.3
Identification marking durability	3.7.6	4.7.6
Group III (2 samples)		
Humidity	3.7.4	4.7.4
Water pressure	3.7.5	4.7.5
Identification marking durability	3.7.6	4.7.6
Impact drop (impact resistance)	3.7.7	4.7.7

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TABLE I. Qualification inspection - Continued.

Inspection	Requirement paragraph	Test method paragraph
Group IV (2 samples)		
Axial cable loading/tension	3.7.8.1	4.7.8.1
Axial cable loading/compression	3.7.8.2	4.7.8.2
Enclosure bending moment	3.7.9	4.7.9
Cable twist	3.7.10	4.7.10
Cable-seal flexing	3.7.11	4.7.11
Mechanical shock	3.7.12	4.7.12
Thermal shock	3.7.13	4.7.13
Operating temperature	3.7.14	4.7.14
Vibration	3.7.15	4.7.15
Salt spray (corrosion)	3.7.16	4.7.16
Flammability	3.7.17	4.7.17

4.4.3 Failures. Any failures shall be cause for refusal to grant qualification.

4.4.4 Retention of qualification. To retain qualification, the contractor shall forward a report at 18-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

a. A summary of the results of the tests performed for inspection of product for delivery (groups A and B), indicating as a minimum the number of lots that have passed, the number that have failed, and the group which they failed. The results of tests of all reworked lots shall be identified and accounted for.

b. A summary of the results of tests performed for periodic inspection (group C), including the number and mode of failures. The summary shall include results of all periodic inspection tests performed and completed during the 18-month period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 18-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 18-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit his qualified products to testing in accordance with the qualification inspection requirements and the reason for no production.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

4.5.1.1 Inspection lot. An inspection lot shall be as defined in MIL-STD-105 and shall consist of all the couplers of the same type, resistance characteristic, and protective enclosure or coating produced under essentially the same conditions and offered for inspection during a period of one month.

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4.5.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table II, and shall be made on the same set of products, in the order shown.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph
Visual and mechanical inspection- - -	3.3, 3.4, 3.5, 3.8, 3.9, 3.10, 3.11, 3.12	---
Transmission matrix - - - - - - - - - -	3.6.6	4.6.6
Identification marking durability - -	3.7.6	4.7.6

4.5.1.2.1 Sampling plan. Tests shall be performed on 100 percent of the product supplied under this specification. Samples that fail group A tests shall not be furnished on contracts. Lots having more than 10 percent total rejects, due to exceeding the specified change limits, shall not be furnished on contracts.

4.5.1.2.1.1 Manufacturer's production inspection. If the manufacturer performs tests similar to those specified in table II as the final step of his production process, group A inspection may be waived and the data resulting from the manufacturer's production tests may be used instead. Authority to waive the inspection shall be granted by the qualifying activity only. The following criteria must be complied with:

a. Tests conducted by the manufacturer during production shall be clearly identical to or more stringent than those specified for group A. Test conditions shall be equal to or more stringent than those specified for group A tests.

b. Manufacturer subjects 100 percent of the product supplied under this specification to his production tests.

c. The parameters measured and the failure criteria shall be the same or more stringent than those specified herein.

d. The lot rejection criteria is the same or more stringent than that specified herein.

e. The manufacturer shall make available all information concerning the test procedures and instrumentation used in his production tests. The manufacturer shall also make available to the Government all records of all detail test data resulting from production tests.

f. Once approved, the manufacturer shall not change the test procedures or criteria without prior notification and concurrence by the qualifying activity.

4.5.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table III in the order shown. They shall be performed on sample units that have been subjected to and have passed the group A inspection.

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TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
Pulse distortion- - - - - - - - - - -	3.6.4	4.6.4
Ambient optical pickup- - - - - - - -	3.6.5	4.6.5
Optical power-angular distribution- -	3.6.7	4.6.7
Thermal shock - - - - - - - - - - - -	3.7.13	4.7.13
Humidity- - - - - - - - - - - - - - -	3.7.4	4.7.4
Axial cable loading		
Tension - - - - - - - - - - - - - -	3.7.8.1	4.7.8.1
Compression - - - - - - - - - - - -	3.7.8.2	4.7.8.2
Cable seal flex - - - - - - - - - - -	3.7.11	4.7.11
Operating temperature - - - - - - - -	3.7.14	4.7.14
Pulse distortion- - - - - - - - - - -	3.6.4	4.6.4
Ambient optical pickup- - - - - - - -	3.6.5	4.6.5
Optical power-angular distribution- -	3.6.7	4.6.7

4.5.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105. Unless otherwise specified herein, S-4 inspection shall be used. The AQL shall be 2.5 percent defective.

4.5.2 Periodic inspection. Periodic inspection shall consist of group C. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.5.2.1.3), delivery of products which have passed groups A and B inspections shall not be delayed pending the results of these periodic inspections.

4.5.2.1 Group C inspection. Group C inspection shall consist of the inspections specified in table IV, in the order shown. Group C inspection shall be performed on sample units of each type and selected from inspection lots which have passed groups A and B inspections. Group C inspection samples shall be representative of production.

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph
Enclosure bending moment- - - - - - -	3.7.9	4.7.9
Mechanical shock- - - - - - - - - - -	3.7.12	4.7.12
Vibration - - - - - - - - - - - - - -	3.7.15	4.7.15
Salt spray (corrosion)- - - - - - - -	3.7.16	4.7.16
Flammability- - - - - - - - - - - - -	3.7.17	4.7.17
Water pressure- - - - - - - - - - - -	3.7.5	4.7.5

4.5.2.1.1 Sampling plan. Every 18 months, 2 sample units which have passed groups A and B inspections shall be selected.

4.5.2.1.2 Disposition of samples. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

4.5.2.1.3 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspections or the inspection which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted; however, final acceptance and shipment shall be

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withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.5.3 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-C-55330.

4.6 Optical test methods. Unless otherwise specified, cladding mode stripping devices shall be used for the output signal. In all cases, a mode stripper device shall be installed in the test circuit between the light source and coupler input. It is imperative that the bandwidth and center wavelength of the detector photodiode match that of the source LED in all sections below. Likewise, the fibers or pigtail fibers for interconnection are to be compatible in terms of diameter, frequency and refractive-index profile characteristics.

4.6.1 Insertion loss (see 3.6.1). Two particular ports are assigned a special status and are selected as the input and output ports on the basis of the below criteria:

a. The specification sheet specifies a particular input port and output port according to proposed application, optical circuit function, and instructions for interconnection to other components; in which case these two ports are considered as the specified or dominant input and output ports.

b. In the absence of the above data, the particular input and output ports shall be chosen as that pair of ports interconnecting the optical channel of least loss, i.e., the dominant path through the coupler device displaying maximum signal transmission. Once determined by means of measurement (or by inspection of the coupler's transmission matrix), this particular pair of ports on the coupler shall be defined as the dominant input and output ports. A suitable optical power source (see figure 1) launches "light" into a specified optical channel containing the coupler under test. The "light" may be static or modulated as appropriate. A cladding mode stripper shall be used to remove cladding modes from the input signal. However, the output power from the specified output port is connected directly to a detector without use of mode stripping. P_J is the total optical signal power into the coupling device under test, and P_I is the total optical signal power out of the coupler under test. The insertion loss in dB is $10 \log_{10} (P_J/P_I)$. P_J and P_I are to be determined just to either side of the device under test so that, if applicable, only the normal coupling losses due to the coupler itself are included. Corrections must be made for all losses (i.e., accessory or interconnection losses) along the optical channel between the points where the optical powers are actually measured, and the points where P_J and P_I are to be determined. Since the source, detector, optical waveguide and interconnection characteristics of the optical channel may change slightly during the course of the test, some control or compensation is required to insure that the optical loss calculated is due solely to the insertion of the device under test. This may be accomplished by proving the stability of the optical channel by independent means, or by using feedback data from a reference optical transmission path to correct the data obtained from the optical channel containing the coupler device under test.

4.6.2 Signal discontinuity (see 3.6.2). The optical ports of the coupler under test (see figure 2) are connected through short optical test cables to an appropriate optical signal source and detector. Mode stripping shall be employed at both the input and output port connections. Unused ports are appropriately capped. The light source provides a static signal easily detected by the detector. The output of the detector is monitored for discontinuities while the coupler under test is subjected to a mechanical or environmental test exposure. Unless otherwise specified (see 3.1), the time interval over which the detector checks for possible discontinuities of coupler output shall extend throughout the entire period of mechanical or environmental exposure. The detector and monitoring equipment shall possess sensitivity and high frequency response to detect discontinuities in the optical signal. The monitoring equipment should include suitable transient capture capability (such as provided by an electronic counter with an adjustable trigger, or oscilloscope with an adjustable sweep trigger and oscilloscope camera).

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4.6.3 Analog modulation (see 3.6.3). The optical ports of the coupler under test (see figure 2) are connected through short optical cables to an optical signal source and detector. Unused ports are appropriately capped. Cladding mode stripping shall be used between the input port (device) and the optical source, and with cladding mode stripping at the output port. The source produces a static signal easily sensed by the detector. The output of the detector is monitored for analog modulation while the optical coupler under test is subjected to a mechanical or environmental test exposure. The detector and monitoring equipment shall possess sensitivity and frequency response to discern analog modulation at the acceptance level. The monitoring equipment should include suitable signal recording capability (such as provided by a triggerable oscilloscope and an oscilloscope camera).

4.6.4 Pulse distortion (see 3.6.4). The optical ports of the coupler under test (see figure 2) are connected through short optical test cables to an optical signal source and detector. In addition to mode stripping the source and the device, mode stripping is used at the relevant output port between the device and the detector. Unused device ports are appropriately capped. The signal source is driven so as to provide a series of optical pulses having maximum rise times and fall times of 1 nanosecond, and uniform pulse durations of approximately 5 nanoseconds. The rise and fall times are measured between the 10 percent and 90 percent optical power levels, and the duration between the 50 percent levels full wave half maximum (FWHM) power. The transmitted optical pulses are compared with the incident optical pulses. The detector and monitoring equipment shall possess sensitivity and frequency response to show both the incident pulses and the transmitted pulses for subsequent comparisons. The monitoring equipment shall also include recording capability (such as provided by an oscilloscope and oscilloscope camera) so that a permanent record of the test data can be obtained. The data acquisition system shall remain in the active monitoring or recording mode over the entire time period of the test exposure.

4.6.5 Ambient optical pickup (see 3.6.5). The optical ports of the coupler under test are either capped or connected to appropriate short optical test cables. The far ends of these cables are either capped or connected to suitable optical power monitoring equipment. The experiment shall be performed both with and without mode stripping at the output port connection. The coupler under test is (as far as practicable) isotropically and homogenously bathed in simulated sunlight in accordance with method 505 of MIL-STD-810. The "light" is broad spectrum with infrared, visible, and ultraviolet components (see method 505 of MIL-STD-810), and illuminates the device under test with an irradiance (power density) of 112 ±5 milliwatts per square centimeter. "Light" emanating from each optical port of the coupler under test shall be measured by the optical power monitoring equipment (having a broad spectral response compatible with the source).

4.6.6 Transmission matrix (see 3.6.6). Each matrix element P_{IJ} represents insertion loss from port J to port I. Thus, each matrix element shall be determined by launching "light" of known power level into port J of the coupler under test and measuring the power level out of port I (see figure 3). The power is measured at all output ports I for each input port J in successive tests until all ports have been individually connected as input ports even for the case where J = I. The "light", from a source, may be either static or modulated as appropriate. Cladding mode stripping of the input power is required. However, there shall be no cladding mode stripping of the output power signals. The transmission matrix elements include coupler connection losses. Tolerances may be associated with each of the specified matrix elements as specified (see 3.1).

4.6.7 Optical power-angular distribution (comparison testing) (see 3.6.7). Matching of input and output angular patterns is described in terms of either method A or method B, as the concept of matching relates to comparing the data obtained from two independent measurements made as required by 4.6.7.1 and 4.6.7.2. Whichever method (A or B) is used in 4.6.7.1, the same method (A or B) shall be used in 4.6.7.2.

a. Method A. The input and output profiles of light energy entering and exiting a coupler shall be measured in accordance with method 6030 (procedure I) of DOD-STD-1678. Procedure I of method 6030 shall be followed along with the method and modifications of 4.6.7.1 (for input pattern) or 4.6.7.2 (for output pattern). The extent of correlation between two patterns shall be defined by comparing the energy registered by the detector at equivalent positions of the input and output patterns: One energy reading with the optical fiber in the goniometer and a corresponding energy reading with the coupler in the goniometer.

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(1) Such localized mapping of the total angular radiation pattern should be done with a pinhole aperture about the detector element so as to minimize the effects of stray reflections. The external noise of room lighting or sunlight must be removed or filtered in the frequency response range of the infrared detector.

(2) The localized output of the coupler, as recorded at each position of the mobile element shall not vary by more than 10 percent from that local signal strength measured for the detector at the same corresponding position of the input pattern (i.e., with no coupler in situ).

b. Method B. The input angular pattern is considered to be the output distribution from a sample length of fiber situated in the same position that the coupler shall occupy.

(1) Alternately, a CCD or CID (charge coupled or injected device) array can be used for a television camera sensor mounted perpendicular to the fiber/coupler axis. This scenario uses a 64 x 64 pixel array to detect and record relative light intensities P as a function of discrete cartesian coordinates (x_i, y_j) at a normal cross-section to the solid angle section encompassing 90 percent of the optical output energy.

(2) The electronic control circuits for the semiconductor array will shift the data into a memory device for data manipulation/comparison after sufficient light exposure has allowed maximum charge accumulation without saturation. Computer hardware and software shall compare the normalized input and output light patterns on a corresponding "pixel by pixel" basis for identical source - CID detector element separation distances. The computer programming shall employ means to notify the user of any particular output pixels which deviate by more than 10 percent from the signal strength registered by the corresponding input pixels, notwithstanding any error tolerance specified by the CCD camera manufacturer.

In the event that the resolution of the 64 x 64 pixel array offers such fine detail as to result in excess sensitivity to slight shifts of alinement; or a noise background causes small signal changes between two corresponding measurements of identical patterns, the computer software shall be modified to compare sum signals from corresponding macro-pixels (groups of many adjacent pixels). The external noise of sunlight or room lighting must be removed or filtered in the frequency response range of the CCD or CID.

4.6.7.1 Input pattern measurement (see 3.6.7.1). The angular radiation pattern supplied (see figure 4) to the coupler at a dominant input port is assumed to equal the far field output from a reference fiber illuminated by an appropriate LED source. The angular profile of this reference fiber output shall be measured in accordance with method 6030 (procedure I) of DOD-STD-1678. The type, size and properties of the optical fiber used as reference fiber shall conform to DOD-C-85045. The reference fiber shall be a straight segment of cable with length of the order of the coupler length. The type of radiation source (together with its center wavelength and spectral bandwidth) to be used shall be chosen to conform with DOD-S-24622. The focusing optics, collimation and mode-stripping preparations used shall be such as to implement total modal injection of the reference fiber input end.

a. Method A. In the event that method A is used, the following modifications and details shall be observed. The output light distribution from the reference fiber will be monitored by a mobile (circular) detector element and recorded as a function of angular position throughout the test.

(1) The angular interval between successive and adjacent positions of the detector during energy measurement shall be 1/2 degree. The range of measurement across the angular profile of reference fiber output shall extend to where the light intensity falls to levels low enough that the detector output drops below twice its noise bias.

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(2) Two full sweeps shall be made across and through the center of the cross-sectional area of the fiber output pattern. The second sweep to be done after the fiber is rotated 90 degrees clockwise from that angular orientation used in the first detector scan. The distance of the detector from the fiber end placed at center of goniometer's sphere shall be any convenient value between 10 to 21 cm. The diameter of the detector element shall be small enough that 1/2 degree increments in angle do not result in any overlap of area between adjacent positions of the detector. For the limits stated on detector distance from fiber end, this condition implies a range on the maximum detector diameter of 0.87 to 1.83 mm. The output from the reference fiber shall be cladding mode stripped of (incoherent) cladding modes before reaching the moveable detector (see figure 4).

b. Method B. In the event that method B is employed, the CCD array shall be placed perpendicular to the source-fiber axis at a distance of 10 to 21 cm from the fiber end. Within these limits, which are sufficient for observation of far-field patterns, the CCD array shall be positioned such that the angular profile covers maximum area on the array. The output from the reference fiber shall be cladding mode-stripped of (incoherent) cladding modes before reaching the CCD array detector (see figure 4). Data from the CCD array or camera corresponding to the reference fiber output pattern shall be stored in electronic memory as image.

$$P_{ij} = P_f \quad (X_i\ Y_j).$$

Note: Whatever equipment (source, detector), accessories, geometries, alinement methods or special techniques are used with the fiber in place will be used during measurements made with the coupler in place.

4.6.7.2 Coupler radiation pattern measurement (see 3.6.7.2). The angular radiation pattern of the coupler output from a dominant output port shall be measured in accordance with method 6030 (procedure I) of DOD-STD-1678. The type of radiation source (together with center wavelength and spectral bandwidth) to be used shall be as specified in DOD-S-24622. The optical-source focusing elements, launch optics (see figure 5), collimation and other components or preparations used shall be such as to implement total modal injection of the same reference fiber used in 4.6.7.1 to characterize an input energy profile for the coupler. A low-loss splice or connector shall be used to join the reference fiber with the dominant input port or pigtail fiber of the coupler in a "straight-line path" parallel to the coupler axis. Hence, the reference fiber becomes an intermediary optical instrument which injects the coupler. Except for the specific input port to be injected and specific output port to be monitored, all ports (or pigtail fibers) of the coupler under test shall be capped with opaque "LED light-proof" covers with light input.

a. Method A. In the event that method A is used, the following modifications and details shall be observed. The output light distribution from the coupler output port (or pigtail fiber) will be monitored by a mobile (circular) detector element and recorded as a function of angular position throughout the test. The angular interval between successive and adjacent positions of the detector during energy measurement shall be 1/2 degree. The range of measurement across the angular profile of coupler output shall extend to where the light intensity falls to levels low enough that the detector output drops below twice its electrical noise bias.

(1) Two full sweeps shall be made across and through the center of the cross-sectional area of the coupler output pattern. The second sweep to be done after the LED source is rotated 90 degrees clockwise from that angular orientation used in the first detector scan. The distance of the detector from the coupler output port at center of goniometers sphere shall be any convenient value between 10 to 21 cm. The diameter of the detector element shall be small enough that 1/2 degree increments in angle do not result in any overlap of area between adjacent positions of the detector. For the limits stated on detector distance from coupler, this condition implies limits upon the maximum detector diameter of 0.87 to 1.83 mm.

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(2) The output from the dominant coupler port shall be cladding mode-stripped of (incoherent) cladding modes before reaching the moveable detector (see figure 5).

b. Method B. In the event that method B is employed, the CCD array shall be placed perpendicular to the source-coupler axis at a distance of 10 to 21 cm from the coupler output port (or pigtail fiber). Within these limits, which are sufficient for observation of far-field patterns, the CCD array shall be positioned such that the angular profile covers maximum area on the array. The output from the coupler shall be cladding mode-stripped of (incoherent) cladding modes before reaching the CCD array detector (see figure 5). Data from the CCD array or camera corresponding to the coupler output pattern shall be stored in electronic memory as image.

$$P_{ij} = P_c\ (X_i, Y_j)$$

Where: $i = 1, 2, 3 \ldots 64$
$j = 1, 2, 3 \ldots 64$
P_f = Relative light intensity at a small area interval ΔX, ΔY in the output from the source fiber.
P_c = Relative light intensity at a small discrete area ΔX, ΔY in the output of the coupler.

Note: Whatever equipment (source or detector), accessories, geometries, alinement methods or special techniques were used with the reference fiber in place must also be used during measurements made with the coupler in place.

4.7 Environmental test methods.

4.7.1 Accelerated aging (see 3.7.1). Fully assembled coupler specimens shall be tested in accordance with test condition 4, storage temperature shall range from -55°C to +105°C, test condition 2, and test time condition B of method 1005 of MIL-STD-1344 unless otherwise specified (see 3.1). The following special test conditions and modifications to method 1005 shall apply to these tests:

a. The coupler shall be exposed to dry air at the proper temperature, after which sufficient time should elapse for the coupler to attain thermal equilibrium at room temperature before optical testing.

b. Pretest and posttest measurements of the coupler dimensions shall be made and reported.

c. Visual inspection of the coupler shall be made using three power magnification.

4.7.2 Ozone exposure (see 3.7.2). Mated cable-coupler assemblies shall be tested in accordance with method 1007 of MIL-STD-1344. After test exposure, the assemblies shall be inspected for damage.

4.7.3 Fluid immersion (see 3.7.3). The cable-coupler assemblies shall be tested in accordance with method 1016 of MIL-STD-1344 with the following modifications. Each assembly shall be tested in a different immersion fluid. Fluids selected for testing shall be the hydraulic fluid type as specified in MIL-L-17331, and the lubricating oil types as specified in MIL-L-17331 or MIL-L-23699. After each test exposure, the assemblies shall be inspected externally for visible damage. After the final exposure, the coupler shall also be internally inspected for fluid penetration.

4.7.4 Humidity (see 3.7.4). Coupler assemblies shall be tested in accordance with the type II, temperature cycling, procedure of method 1002 of MIL-STD-1344. Cleaning of the unmated coupler-connector junctions prior to optical testing is permissible, if necessary. The following special test conditions and modifications to method 1002 apply.

a. Pretest and posttest measurements of the coupler-dimensions and coupler-cable outer diameter shall be made and reported.

b. Visual inspection of the specimen shall be made using three power magnification.

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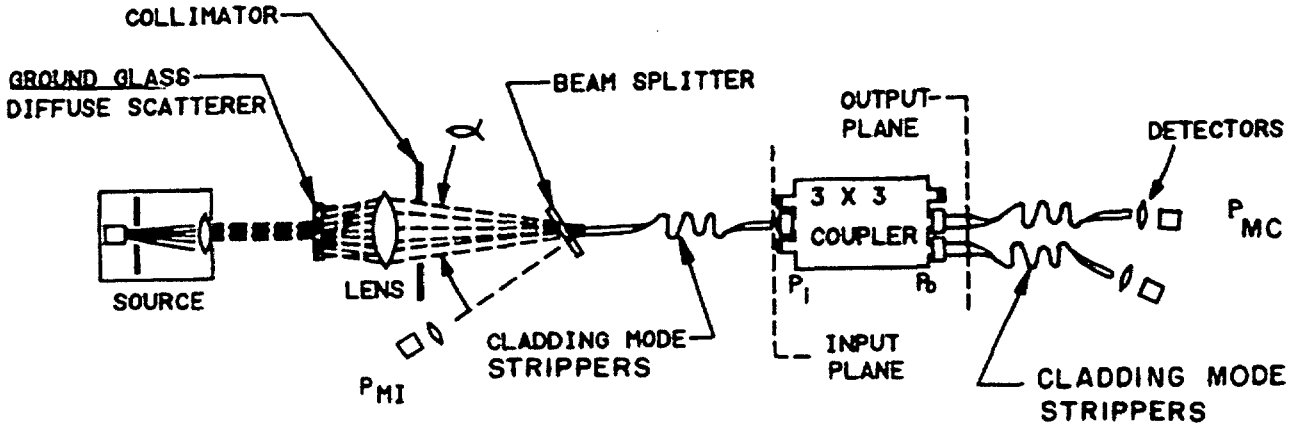


FIGURE 1. Type I coupler total injection configuration.

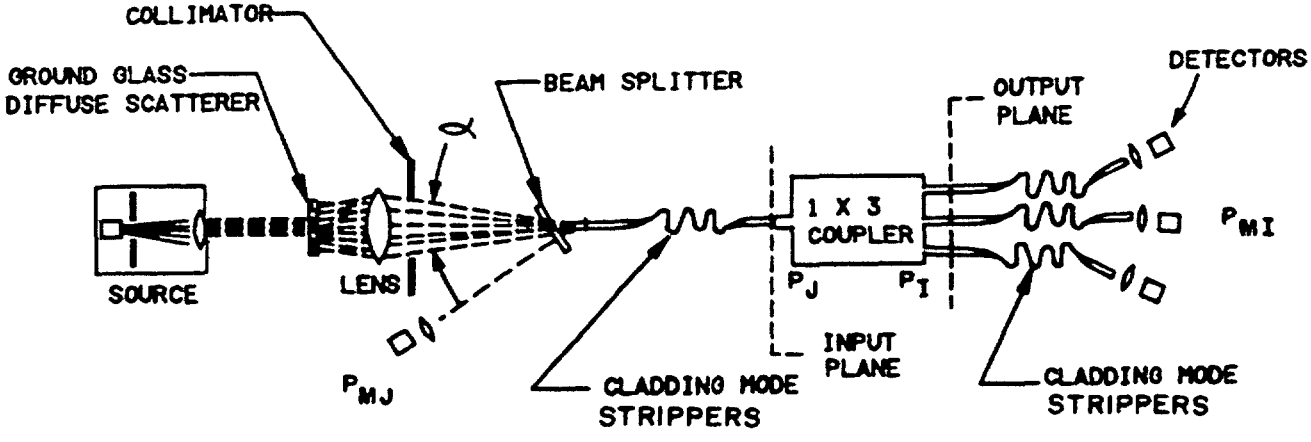


FIGURE 2. Discontinuity, analog modulation and pulse distortion test configuration (with cladding mode stripper).

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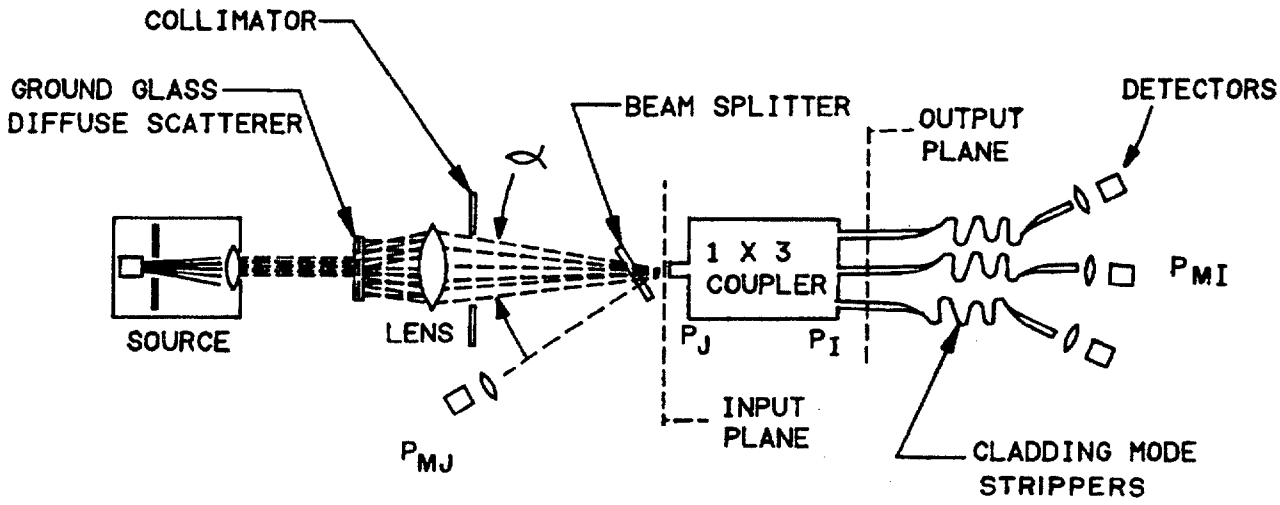


FIGURE 3. Insertion loss and transmission matrix test configuration with cladding mode stripping.

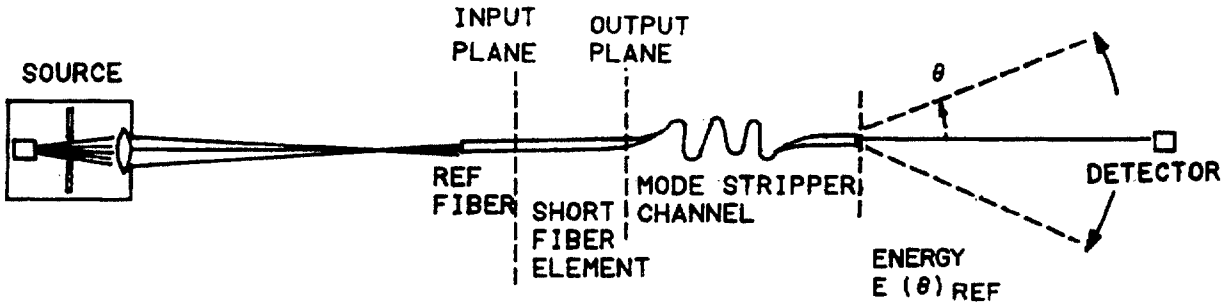
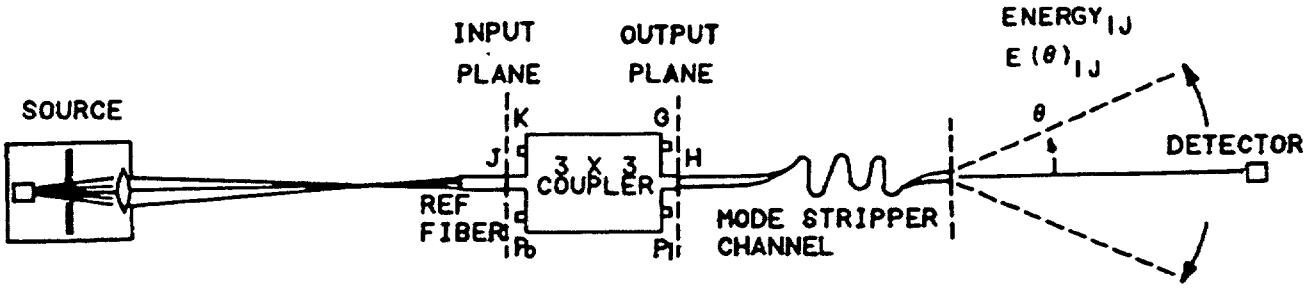


FIGURE 4. Input radiation pattern (for reference fiber).

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NOTES:
1. A short length of fiber is added to extend the plane at which mode injection of reference system occurs to the same corresponding position as that for which modal injection of coupler system occurs below.
2. Mobil scanning energy detector is mounted in pivotable goniometer.

FIGURE 5. Output radiation pattern (coupler output) for dominant port-to-port path through coupler characterized by P_{IJ}.

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4.7.5 Water pressure (see 3.7.5). Mated coupler-cable assemblies shall be tested for water pressure susceptibility as follows. The assemblies shall be immersed in an aqueous dye penetrant solution to a depth of 2.0 meters for a period of 48 hours. The solution temperature shall be maintained between 10°C and 35°C during the exposure period. The dye concentration shall visibly indicate liquid penetration. After test exposure, the coupler assemblies shall be externally cleaned, externally inspected, mated, optically tested, and then internally inspected for dye penetration into the coupler.

4.7.6 Identification marking durability (see 3.7.6). The marking durability test shall be accomplished as described herein. The marking durability tester shall be designed to firmly hold the coupler with the marked face fully exposed. The tester shall be capable of rubbing a small cylindrical steel mandrel, 1.0 ±0.1 millimeter in diameter, repeatedly over the exposed surface of the coupler, as held in such a position that the longitudinal axes of the mandrel and the markings are at right angles to each other with their surfaces in contact. One kilogram mass affixed to a jig above the mandrel shall control the force exerted normal to the marking on exposed surface of the coupler. A motor driven mechanism and counter shall be used to deliver an accurate number of abrading strokes in a direction parallel to the direction of the marking. The length of the stroke shall be 10 millimeters and the frequency shall be 120 strokes (60 stroking cycles) per minute. The coupler shall be subjected to 600 strokes (300 stroking cycles).

4.7.7 Impact drop (impact resistance) (see 3.7.7). Unmated coupler assemblies shall be tested in accordance with the following method. A total of three drops on each assembly shall be made from a height of one meter above a flat solid steel plate of thickness not less than .50 inch. Each fall shall be oriented such that a coupler surface impacts the floor approximately normal to a different principal axis of the coupler geometry. Three different faces suffer impacts in the three successive and different drop tests where the coupler body is rotated by an angle of 90 degrees relative to its reference (initial) position between each free-fall. The coupler shall be set into free-fall motion without regard to careful angular alinement before release, and without regard to possible angular shifts of orientation after release. This test accounts for the situation where a coupler is allowed to drop onto its connector outlets, pigtails, plugs, or receptacles. However, all such "high-risk" drop shall be done with protective caps in-place to cover the vulnerable connector areas on type I couplers. The coupler shall then be dropped onto sides or surfaces bearing connector outlets wherever possible in each of the drop tests.

4.7.8 Axial cable loading.

4.7.8.1 Axial cable loading/tension (see 3.7.8.1). The axial cable loading test will be performed according to the procedure described herein.

a. The coupler shall be securely mounted in the normal manner.

b. All cables on the most populated face shall be mechanically loaded in tension to one-sixth of the maximum tensile load for the pigtail cable.

c. Apply the tensile load at the rate of about 2000 newtons per minute, and hold the maximum load for one minute.

d. Remove the tensile load.

e. Visually inspect the coupler using three power magnification.

4.7.8.2 Axial cable loading/compression (see 3.7.8.2). The axial compressive load test shall be performed as described herein. If all parts of the coupler are essentially the same, only the input and output ports used for the insertion loss measurement need be tested.

a. The test cables forming a part of the assembled coupler shall be gripped over a length equal to at least three cable diameters, starting one cable diameter from the coupler enclosure or the strain relief mechanism if it is external to the enclosure. The coupler enclosure shall be gripped so as not to alter any of its mechanical properties.

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b. The gripped cable and coupler enclosure shall be forced together along the direction of the cable axis where the cable enters the coupler assembly. The force magnitude is a function of pigtail cable diameter as listed in table V. Loading and unloading rates are such that the coupler is compressively loaded and unloaded linearly in time, with the loading and unloading times each less than 10 seconds. The time spent under full compressive load shall be at least one minute. Tensile loads shall not be applied to the pigtail cables.

TABLE V. Force magnitude vs. pigtail cable diameter.

Nominal cable diameter	Compressive axial force (newtons)
0 mm to 3 mm	10
3 mm to 6 mm	20
6 mm to 10 mm	50
1 cm to 2 cm	100
2 cm and larger	200

c. Any test fixture may be used provided it adequately grips the coupler enclosure and cable, allows control and measurement of the applied compressive axial force, and allows visual observation of the region where the cable meets the splice assembly. The test fixture must also allow for the performance of all optical tests called for while the assembly is subject to the compressive axial load.

d. With load applied, measure the distance between the fixture clamp and the enclosure or strain relief mechanism and record the insertion loss.

e. Remove the load.

f. Visual observations and inspection shall be made with three power magnification.

4.7.9 Enclosure bending moment (see 3.7.9). Mated coupler-connector assemblies shall be tested in accordance with the following procedure which checks proper operation of coupler under deformation due to applied torque, a condition which can occur as a result of uneven stresses from mounting or installation. This test considers as a reference, the practical case of a coupler body 50 mm (0.020 inch) in length, mounted on a flat surface made uneven by a .5 mm (20 mm) shim. This small shim, which is used to model a typical convex surface of attachment in shipyard construction, is to be situated midway underneath the longest extent of the coupler body. The coupler shall be balanced in a stable position atop a small metal block (or shim). The width of the fulcrum (a bar of shim), upon which the coupler rests shall be less than or equal to 0.1 mm of the maximum dimension across the top face of the coupler (but never with the contact span less than 2 mm under any circumstances). The fulcrum shall be secured against a smooth, level foundation offering stiff support. Install the coupler according to the manufacturers recommended mounting instructions, but with the shim centered underneath the coupler. In the event that such instructions are not available from the manufacturer, the sample description below pertains to a coupler bearing four screw holes for attachment to a flat base, as illustrated in figure 6A. The four screws are to be tightened gradually in quarter-turn clockwise increments once the coupler body becomes loosely clamped against the fulcrum during this fastening process. The four screws shall be attended by small quarter-turn twists one after another in cyclic permutation such that all four bolts become progressively tighter. The force applied by the mounting screws of the coupler shall be increased until the loading effect is sufficient to deflect the far edges downward to the level of the foundation (but only to the degree that the tips of the coupler body just barely touch the foundation below). This can be determined electronically by sensing the moment that an electrical conduction path is

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established while the coupler gradually becomes more tightly fastened. The insertion loss of the "bent" coupler is measured at this time and compared with the maximum allowed value, as specified (see 3.1). Each of the four screws (fasteners) shall be tightened. The test configuration and definition of bending moment angle θ_{BM} in terms of the deformed coupler body are shown on figure 6B. Note: The combination of the shim and coupler, as shown, defines a reference angle (of deformation) of 1.15 degrees. The insertion loss of the deformed coupler under stress/strain is then measured to verify that the insertion loss (including any-induced attenuation caused by this test) still conforms to values less than the maximum allowed insertion loss, as specified (see 3.1). Note: The fulcrum bar (or shim) shall be of ample length to extend across the full width of the coupler body. Both the pivot piece of shim (fulcrum bar) and floor support beneath the fulcrum shall be of rigid, inelastic and smoothly finished material. The proper height Z (of shim) required for a fulcrum to induce a 1.15 degrees deflection angle on a coupler of length A is (according to the geometry of figure 6B):

$$Z = 1/2\ A \tan (1.15^\circ)$$

$$Z \cong 0.01\ A$$

4.7.10 Cable twist (for type II couplers with pigtails only) (see 3.7.10). Coupler assemblies shall be tested as follows. If all ports of the coupler are essentially the same, this test may be performed on one port only. The coupler-housing shall be clamped in a rotatable fixture capable of ±360 degree rotation. The assembly pigtail cables shall be stretched with minimum tension (force of 45 newtons) to their maximum lengths and clamped to the test fixture at a distance of about 100 times the output cable diameter from the coupler. The fixture shall be rotated ±360 degrees at a rate of one cycle per 5 seconds for a total of 50 cycles. At the completion of the required number of cycles, visually inspect the coupler under three power magnification.

4.7.11 Cable-seal flexing (for type II couplers with pigtails only) (see 3.7.11). Coupler assemblies shall be tested in accordance with method 2017 of MIL-STD-1344. If all seals are essentially identical, only one seal on the input or output cables (pigtail leads) used for the insertion loss test need be flexed.

4.7.12 Mechanical shock (see 3.7.12). Mated coupler cable assemblies shall be tested in accordance with the methods of MIL-S-901, grade A, Class I, Type B.

4.7.13 Thermal shock (see 3.7.13). Thermal shock testing shall be accomplished in accordance with method 4020 of DOD-STD-1678 (see method 107 of MIL-STD-202, as a guideline). Cleaning of optical contacts of coupler connectors prior to the test exposure shall be permitted. The following special test conditions and modifications to method 4020 of DOD-STD-1678 shall apply to these tests:

a. Test method cycles shall be as indicated in test condition A, except that the low and high exposure temperatures are -40°C and +71°C, respectively.

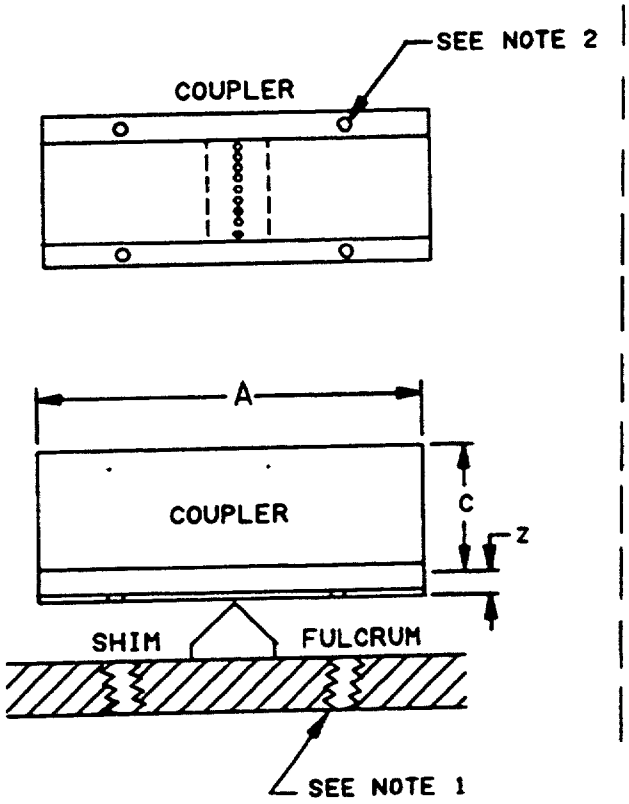
b. The time duration of each half cycle temperature exposure (steps 1 and 3) shall be the minimum (1/2 hour) as specified in method 107 of MIL-STD-202. The maximum time condition (5 minutes) is specified for the interim temperature of steps 2 and 4.

c. Signal continuity shall be monitored during one complete positive temperature transition (from the low temperature limit to the high temperature limit), and one complete negative temperature transition (from the high temperature limit to the low temperature limit). During the last temperature cycle, insertion loss shall be measured at the high temperature limit and the low temperature limit.

d. Pretest and posttest measurements of the couplers dimensions shall be made and reported.

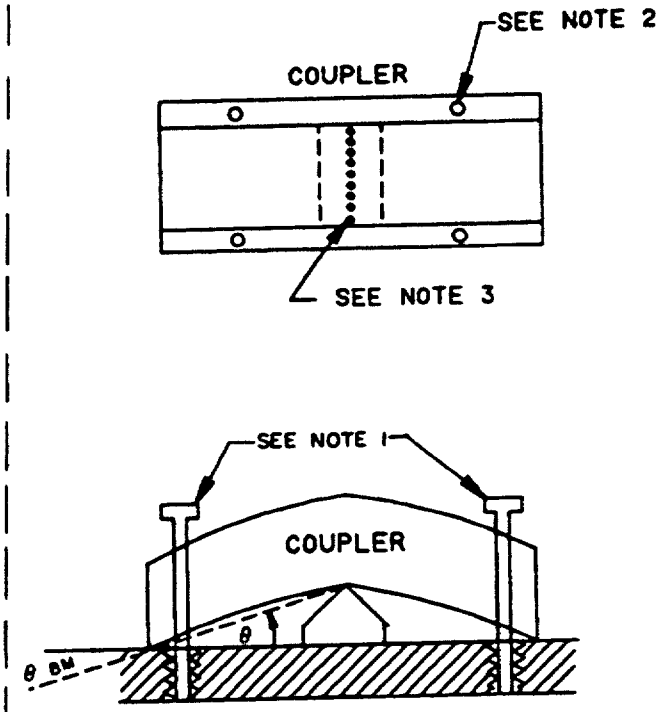
e. Visual inspection of the coupler shall be accomplished using three power magnification.

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NOTES:
1. Predrilled holes to implement flange type mounting of coupler over area with shim.
2. Four preexisting holes in flange edge provided by manufacturer for easy, rigid mounting of coupler body.

FIGURE 6A. Enclosure bending moment test configuration (without loading).



NOTES:
1. Screw-driven torque shall be applied at all four points of mounting via flange.
2. Loading points on coupler after tight fastening.
3. Shim strip or fulcrum "lies" underneath middle zone of coupler.

FIGURE 6B. Enclosure bending moment test configuration (with loading).

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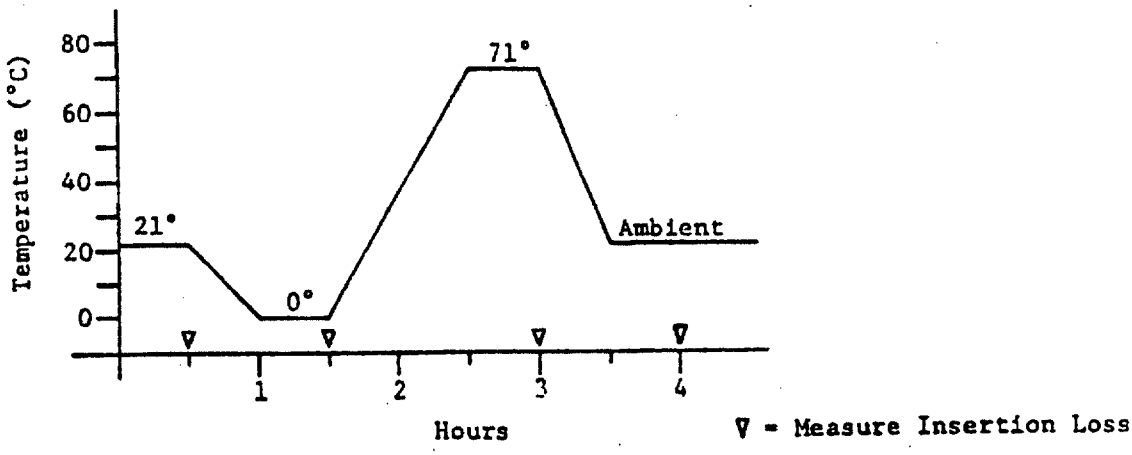


FIGURE 7. Operating temperature range test.

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4.7.14 Operating temperature (see 3.7.14). This test measures the insertion loss of couplers at several temperatures throughout the operating temperature range of 0°C - +71°C. Test chambers shall provide the temperature environment over the specified range. The volume of each chamber shall be of sufficient size so that the test sample will not interfere with the generation and maintenance of test conditions. Any heat source shall be located such that radiant heat will not fall directly on the test sample. Thermocouples or equivalent temperature sensors shall be used to determine and control the temperature within each chamber.

a. The coupler shall be placed in the test chamber. The cables which lead from the dominant input and output ports shall exit the chamber and shall be connected to optical instrumentation for monitoring of insertion loss (see 4.6.1).

b. The test chamber temperature shall be stabilized at 21°C for 30 minutes, insertion loss shall be measured and recorded at the end of this period before the temperature is changed (see figure 7). The temperature of the test chamber shall be lowered to 0°C over a 30 minute period and maintained at that temperature for the next 30 minutes. The insertion loss shall be measured at the end of the 30 minute period while the temperature remains unchanged. The chamber temperature shall be raised to 71°C over a 60 minute period. After the temperature has stabilized at 71°C for 30 minutes and while at that temperature, the insertion loss shall be measured. Decrease the chamber temperature to ambient over a 30 minute period, maintain ambient temperature for an additional 30 minutes. Insertion loss shall be measured at the end of that period.

c. Remove the coupler from the test chamber; visually inspect the coupler using three power magnification and the outside of the coupler for any damage to the seals, connectors or the enclosure.

4.7.15 Vibration (see 3.7.15). Type I couplers with mated cable assemblies shall be tested in accordance with test condition II of method 2005 of MIL-STD-1344.

4.7.16 Salt spray (corrosion) (see 3.7.16). Mated coupler cable assemblies shall be tested in accordance with test condition C of method 1001 of MIL-STD-1344. After test exposure, the assemblies shall be externally cleaned and inspected for evidence of corrosion.

4.7.17 Flammability (see 3.7.17). Flammability of couplers and associated accessories shall be tested under test condition B in accordance with method 1012 of MIL-STD-1344. Flammability retardance of "pigtail-fiber" cables (in the case of type II couplers) shall be tested in accordance with method 5010 of DOD-STD-1678.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-C-55330.

6. NOTES

6.1 Intended use. The fiber optic couplers covered by this specification are intended for use in any application where their performance characteristics are required. The couplers are suitable for installation on all military systems within the limitation of their specified performance requirement.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following.

a. Title, number, and date of this specification.

b. Specification sheet number, title, and date.

c. Part number.

d. External color required (see 3.10).

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e. Exceptions, if any, to the optional provisions of this specification are as follows:

(1) Exceptions to product identification requirements (3.11), if applicable.

(2) Responsibility for inspection, if other than specified (see 4.1).

(3) Special preparation for delivery requirements, if applicable (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the Qualified Products List QPL-24621 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is Naval Sea Systems Command, SEA 55Z3, DOD Standardization Program and Documents Division, Washington, DC 20362; however, information pertaining to qualification of products may be obtained from Defense Electronics Supply Center, DESC-EQ, Dayton, Ohio 45444.

6.3.1 Provisions governing qualification. Copies of "Provisions Governing Qualification" SD-6 may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Definitions. Definitions of terms shall be in accordance with DOD-STD-1678 and as specified herein.

6.4.1 Insertion loss (port-to-port). The insertion loss of an optical coupling device is the optical signal power loss in an optical channel caused by insertion of the optical coupler into the optical circuit containing that optical channel. Insertion loss defines a value for the attenuation of a particular optical path from input port J to output port I through the coupler. The test provides a means of investigating and quantifying possible performance degradation during or after environmental or mechanical test exposures by comparison against pretest values.

6.4.2 Signal discontinuities. A signal discontinuity refers to a relatively sharp, short duration interruption or "drop-out" in the optical signals transmitted by an optical coupling device. The purpose of a continuity test is to verify that the coupling device does not introduce any signal discontinuities when it is subjected to mechanical or environmental stress.

6.4.3 Analog modulation. Analog modulation refers to dynamic amplitude modulation of the optical signals transmitted by an optical coupling device. The purpose of an analog modulation test is to verify that the optical coupler in question does not excessively modulate its transmitted optical signals when it is subjected to mechanical or environmental stress.

6.4.4 Pulse distortion. Pulse distortion refers to a change in the shape or duration of optical pulses transmitted by an optical coupling device. The purpose of a distortion test is to verify that the optical coupler in question does not excessively alter the transmitted optical pulse waveform.

6.4.5 Ambient optical pickup. Ambient optical pickup refers to the coupling of ambient optical power incident upon the exterior of a coupling device into the optical signal channels of the coupling device. The purpose of an ambient optical link test is to verify the immunity of the optical coupling device to ambient radiation (optical interference).

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6.4.6 Transmission matrix. The transmission matrix characterizes the distribution of input energy from an input port "J" among all other ports "I". For an M x N port coupler, the matrix is

$$\begin{matrix} P_{11} & P_{12} & P_{13} & \cdots & P_{1N} \\ P_{21} & P_{22} & \cdots & & \\ \cdots & & & & \\ \cdots & & & & \\ P_{MI} & \cdots & & & P_{MN} \end{matrix}$$

Each matrix element P_{IJ} (in dB) is the optical power level out of port I relative to the optical power level input to port J: $P_{IJ} = 10 \log (P_I/P_J)$. The purpose of a transmission matrix test is to confirm the reputed transmission characteristics of the coupler.

6.4.7 Total injection by an optical source. The term "total injection" is used to denote uniform excitation of all modes capable of propagating without excessive loss through the fiber optic waveguide or device.

6.4.8 Mode-stripper device. Cladding mode stripping refers to a method of eliminating the high-loss modes from a fiber optic input signal, before introduction into the optical component of interest, e.g., detector, coupler or waveguide. This can help to ensure detector measurements that represent practical systems operate at equilibrium levels.

6.4.9 Passive couplers. A passive coupler is an optical device which splits or divides an input signal at one port into two or more output signals (with each output signal extracted from different ports) without any modification, modulation, or amplification of the input within the coupler. A passive coupler does not contain an optical source, detector or photosensitive element of any type.

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Navy - AS, EC
Air Force - 11, 17, 19, 80, 99
DLA - ES

User activities:
Navy - CG, MC

Agent:
DLA - ES

Preparing activity:
Navy - SH

(Project 6060-0021)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-C-24621	Couplers, Passive, Fiber Optic General Specification For (Metric)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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